

Q1 2021 (01 Sep Shift 2)

A man starts walking from the point $P(-3, 4)$ touches the x -axis at R , and then turns to reach at the point $Q(0, 2)$. The man is walking at a constant speed. If the man reaches the point Q in the minimum time, then $50((PR)^2 + (RQ)^2)$ is equal to .

Q2 2021 (01 Sep Shift 2)

Let θ be the acute angle between the tangents to the ellipse $\frac{x^2}{9} + \frac{y^2}{1} = 1$ and the circle $x^2 + y^2 = 3$ at their point of intersection in the first quadrant. Then $\tan \theta$ is equal to :

- (1) $\frac{5}{2\sqrt{3}}$
- (2) $\frac{2}{\sqrt{3}}$
- (3) $\frac{4}{\sqrt{3}}$
- (4) 2

Q3 2021 (01 Sep Shift 2)

The function $f(x) = x^3 - 6x^2 + ax + b$ is such that

$f(2) = f(4) = 0$. Consider two statements.

(S1) there exists $x_1, x_2 \in (2, 4)$, $x_1 < x_2$, such that

$$f'(x_1) = -1 \text{ and } f'(x_2) = 0$$

(S2) there exists $x_3, x_4 \in (2, 4)$, $x_3 < x_4$, such that

f is decreasing in $(2, x_4)$, increasing in $(x_4, 4)$

$$\text{and } 2f'(x_3) = \sqrt{3}f(x_4)$$

Then

- (1) both (S1) and (S2) are true
- (2) (S1) is false and (S2) is true
- (3) both (S1) and (S2) are false
- (4) (S1) is true and (S2) is false

Q4 2021 (31 Aug Shift 2)

Let $f(x)$ be a cubic polynomial with $f(1) = -10$, $f(-1) = 6$, and has a local minima at $x = 1$, and $f'(x)$ has a local minima at $x = -1$. Then $f(3)$ is equal to .

Q5 2021 (31 Aug Shift 2)

Let f be any continuous function on $[0, 2]$ and twice differentiable on $(0, 2)$. If $f(0) = 0$, $f(1) = 1$

and $f(2) = 2$, then

(1) $f''(x) = 0$ for all $x \in (0, 2)$

(2) $f''(x) = 0$ for some $x \in (0, 2)$

(3) $f'(x) = 0$ for some $x \in [0, 2]$

(4) $f''(x) > 0$ for all $x \in (0, 2)$

Q6 2021 (31 Aug Shift 1)

If ' R ' is the least value of ' a ' such that the function $f(x) = x^2 + ax + 1$ is increasing on $[1, 2]$ and ' S ' is the greatest value of ' a ' such that the function $f(x) = x^2 + ax + 1$ is decreasing on $[1, 2]$, then the value of $|R - S|$ is .

Q7 2021 (27 Aug Shift 2)

A box open from top is made from a rectangular sheet of dimension $a \times b$ by cutting squares each of side x from each of the four corners and folding up the flaps. If the volume of the box is maximum, then x is equal to :

(1) $\frac{a+b-\sqrt{a^2+b^2-ab}}{12}$

(2) $\frac{a+b-\sqrt{a^2+b^2+ab}}{6}$

(3) $\frac{a+b-\sqrt{a^2+b^2-ab}}{6}$

(4) $\frac{a+b+\sqrt{a^2+b^2-ab}}{6}$

Q8 2021 (27 Aug Shift 1)

A wire of length 20 m is to be cut into two pieces.

One of the pieces is to be made into a square and the other into a regular hexagon. Then the length of the side (in meters) of the hexagon, so that the combined area of the square and the hexagon is minimum, is:

(1) $\frac{5}{2+\sqrt{3}}$

(2) $\frac{10}{2+3\sqrt{3}}$

(3) $\frac{5}{3+\sqrt{3}}$

(4) $\frac{10}{3+2\sqrt{3}}$

Q9 2021 (26 Aug Shift 2)

The local maximum value of the function

$$f(x) = \left(\frac{2}{x}\right)^{x^2}, x > 0, \text{ is}$$

(1) $(2\sqrt{e})^{\frac{1}{e}}$

(2) $\left(\frac{4}{\sqrt{e}}\right)^{\frac{e}{4}}$

(3) $(e)^{\frac{2}{e}}$

(4) 1

Q10 2021 (26 Aug Shift 1)

A wire of length 36 m is cut into two pieces, one of the pieces is bent to form a square and the other is bent to form a circle. If the sum of the areas of the two figures is minimum, and the circumference of the circle is k (meter), then $\left(\frac{4}{\pi} + 1\right)k$ is equal to .

Answer Key

Q1 (1250)

Q2 (2)

Q3 (1)

Q4 (22)

Q5 (2)

Q6 (2)

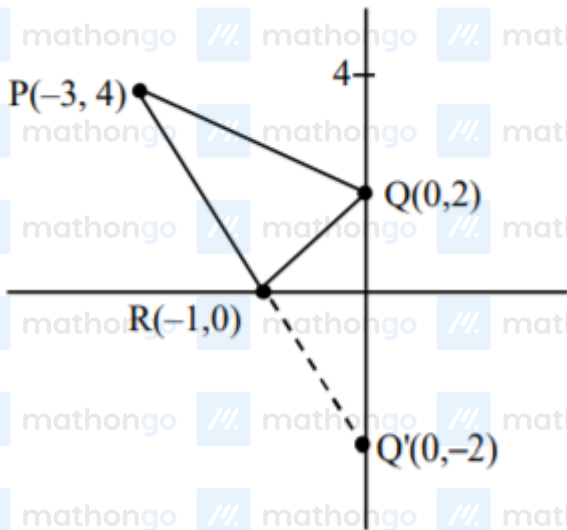
Q7 (3)

Q8 (4)

Q9 (3)

Q10 (36)

Q1 (1250)



$$50 (PR^2 + RQ^2)$$

$$50(20 + 5)$$

$$50(25)$$

$$= 1250$$

Q2 (2)

The point of intersection of the curves $\frac{x^2}{9} + \frac{y^2}{1} = 1$ and $x^2 + y^2 = 3$ in the first quadrant is

$$\left(\frac{3}{2}, \frac{\sqrt{3}}{2}\right)$$

Now slope of tangent to the ellipse $\frac{x^2}{9} + \frac{y^2}{1} = 1$ at $\left(\frac{3}{2}, \frac{\sqrt{3}}{2}\right)$ is

$$m_1 = -\frac{1}{3\sqrt{3}}$$

And slope of tangent to the circle at $\left(\frac{3}{2}, \frac{\sqrt{3}}{2}\right)$ is m_2

$$= -\sqrt{3}$$

So, if angle between both curves is θ then

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right| = \left| \frac{-\frac{1}{3\sqrt{3}} + \sqrt{3}}{1 + \left(-\frac{1}{3\sqrt{3}}\right)(-\sqrt{3})} \right|$$

$$= \frac{2}{\sqrt{3}}$$

Option (2)

Q3 (1)

$$f(x) = x^3 - 6x^2 + ax + b$$

$$f(2) = 8 - 24 + 2a + b = 0$$

$$2a + b = 16 \dots (1)$$

$$f(4) = 64 - 96 + 4a + b = 0$$

$$4a + b = 32 \dots (2)$$

Solving (1) and (2)

$$a = 8, b = 0$$

$$f(x) = x^3 - 6x^2 + 8x$$

$$f(x) = x^3 - 6x^2 + 8x$$

$$f'(x) = 3x^2 - 12x + 8$$

$$f''(x) = 6x - 12$$

 $\Rightarrow f'(x)$ is $\uparrow$ for $x > 2$, and $f'(x)$ is $\downarrow$ for $x < 2$

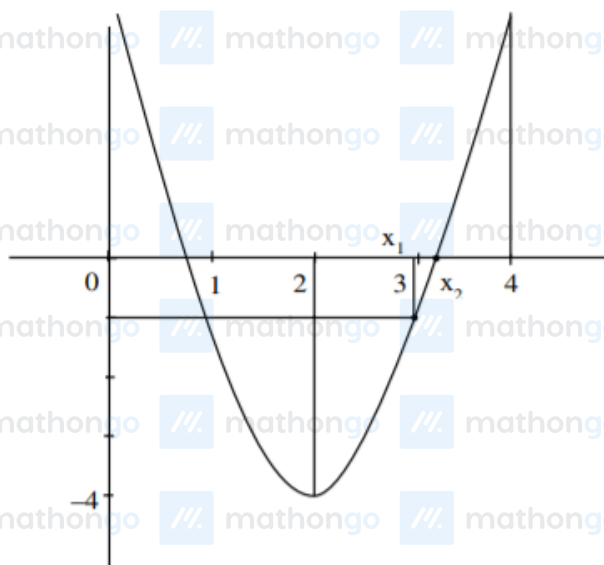
$$f'(2) = 12 - 24 + 8 = -4$$

$$f'(4) = 48 - 48 + 8 = 8$$

$$f'(x) = 3x^2 - 12x + 8$$

vertex $(2, -4)$

$$f'(2) = -4, f'(4) = 8, f'(3) = 27 - 36 + 8$$



$$f'(x_1) = -1, \text{ then } x_1 = 3$$

$$f'(x_2) = 0$$

Again

$$f'(x) < 0 \text{ for } x \in (2, x_4)$$

$$f'(x) > 0 \text{ for } x \in (x_4, 4)$$

$$x_4 \in (3, 4)$$

$$f(x) = x^3 - 6x^2 + 8x$$

$$f(3) = 27 - 54 + 24 = -3$$

$$f(4) = 64 - 96 + 32 = 0$$

$$\text{For } x_4(3, 4)$$

$$f(x_4) < -3\sqrt{3}$$

$$\text{and } f'(x_3) > -4$$

$$2f'(x_3) > -8$$

$$\text{So, } 2f'(x_3) = \sqrt{3}f(x_4)$$

$$\text{Correct Ans. (1)}$$

Q4 (22)

$$F'(x) = a(x-1)(x+3)$$

$$F''(x) = 6a(x+1)$$

$$F'(x) = 3a(x+1)^2 + b$$

$$F'(1) = 0 \Rightarrow b = -12a$$

$$F(x) = a(x+1)^3 - 12ax + c$$

$$= (x+1)^3 - 12x - 6$$

$$F(3) = 64 - 36 - 6 = 22$$

Q5 (2)

$$f(0) = 0, f(1) = 1 \text{ and } f(2) = 2$$

Let $h(x) = f(x) - x$ has three roots

By Rolle's theorem $h'(x) = f'(x) - 1$ has at least two roots

$h''(x) = f''(x) = 0$ has at least one roots

Q6 (2)

$$f(x) = x^2 + ax + 1$$

$$f'(x) = 2x + a$$

when $f(x)$ is increasing on $[1, 2]$

$$2x + a \geq 0 \quad \forall x \in [1, 2]$$

$$a \geq -2x \quad \forall x \in [1, 2]$$

$$R = -4$$

when $f(x)$ is decreasing on $[1, 2]$

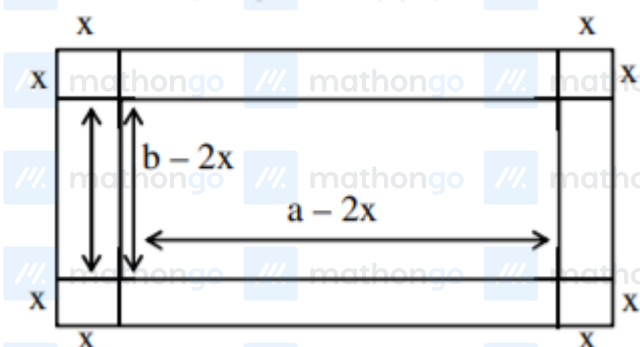
$$2x + a \leq 0 \quad \forall x \in [1, 2]$$

$$a \leq -2 \quad \forall x \in [1, 2]$$

$$S = -2$$

$$|R - S| = |-4 + 2| = 2$$

Q7 (3)



$$V = \ell. b. h = (a - 2x)(b - 2x)x$$

$$\Rightarrow V(x) = (2x - a)(2x - b)x$$

$$\Rightarrow V(x) = 4x^3 - 2(a + b)x^2 + abx$$

$$\Rightarrow \frac{d}{dx}v(x) = 12x^2 - 4(a + b)x + ab$$

$$\frac{d}{dx}(v(x)) = 0 \Rightarrow 12x^2 - 4(a + b)x + ab = 0 < \frac{\alpha}{\beta}$$

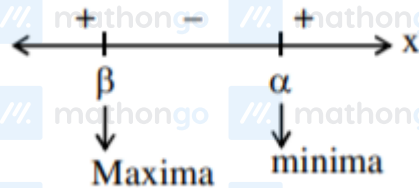
$$\Rightarrow x = \frac{4(a+b) \pm \sqrt{16(a+b)^2 - 48ab}}{2(12)}$$

$$= \frac{(a+b) \pm \sqrt{a^2 + b^2 - ab}}{6}$$

$$\text{Let } x = \alpha = \frac{(a+b) + \sqrt{a^2 + b^2 - ab}}{6}$$

$$\beta = \frac{(a+b) - \sqrt{a^2 + b^2 - ab}}{6}$$

$$\text{Now, } 12(x - \alpha)(x - \beta) = 0$$

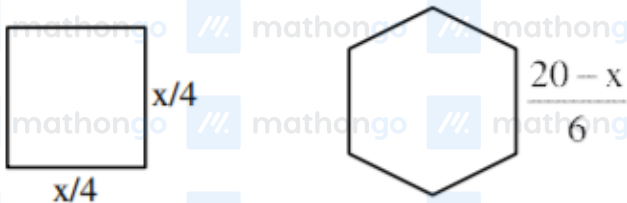


$$\therefore x = \beta$$

$$= \frac{a+b - \sqrt{a^2 + b^2 - ab}}{6}$$

Q8 (4)

Let the wire is cut into two pieces of length x and $20 - x$.



$$\text{Area of square} = \left(\frac{x}{4}\right)^2 \quad \text{Area of regular hexagon} = 6 \times \frac{\sqrt{3}}{4} \left(\frac{20-x}{6}\right)^2$$

$$\text{Total area} = A(x) = \frac{x^2}{16} + \frac{3\sqrt{3}}{2} \frac{(20-x)^2}{36}$$

$$A'(x) = \frac{2x}{16} + \frac{3\sqrt{3} \times 2}{2 \times 36} (20-x)(-1)$$

$$A'(x) = 0 \text{ at } x = \frac{40\sqrt{3}}{3+2\sqrt{3}}$$

$$\text{Length of side of regular Hexagon} = \frac{1}{6}(20-x)$$

$$= \frac{1}{6} \left(20 - \frac{40\sqrt{3}}{3+2\sqrt{3}} \right)$$

$$= \frac{10}{2+2\sqrt{3}}$$

Q9 (3)

$$f(x) = \left(\frac{2}{x}\right)^{x^2}; x > 0$$

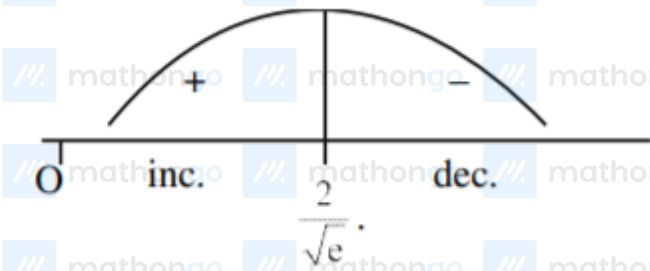
$$\ln f(x) = x^2(\ln 2 - \ln x)$$

$$f'(x) = f(x)\{-x + (\ln 2 - \ln x)2x\}$$

$$f'(x) = \underbrace{f(x)}_{+} \underbrace{x}_{+} \underbrace{(2\ln 2 - 2\ln x - 1)}_{g(x)}$$

$$g(x) = 2\ln 2 - 2\ln x - 1$$

$$= \ln \frac{4}{x^2} - 1 = 0 \Rightarrow x = \frac{2}{\sqrt{e}}$$



$$LM = \frac{2}{\sqrt{e}}$$

$$\text{Local maximum value} = \left(\frac{2}{2/\sqrt{e}}\right)^{\frac{4}{e}} \Rightarrow e^{\frac{2}{e}}$$

Q10 (36)

$$\text{Let } x + y = 36$$

x is perimeter of square and y is perimeter of circle side of square = $x/4$

$$\text{radius of circle} = \frac{y}{2\pi}$$

$$\text{Sum Areas} = \left(\frac{x}{4}\right)^2 + \pi\left(\frac{y}{2\pi}\right)^2$$

$$= \frac{x^2}{16} + \frac{(36-x)^2}{4\pi}$$

For min Area :

$$x = \frac{144}{\pi+4}$$

$$\Rightarrow \text{Radius} = y = 36 - \frac{144}{\pi+4}$$

$$\Rightarrow k = \frac{36\pi}{\pi+4}$$

$$\left(\frac{4}{\pi} + 1\right)k = 36$$

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